Name: _____

Chapter 5 Practice Quiz: Newton's Third Law and Vectors

Part 1: Choose the best choice

1. A soccer player kicks a ball. The action force is "the foot pushes on the ball." What is the reaction force? **[C]**

   A) Earth pulls down on the ball.
   B) The ball speeds up and moves forward.
   C) The ball pushes on the player's foot.
   D) The ground pushes up on the player.

2. You push on a wall with a force of 50 N. How hard does the wall push on you? **[C]**

   A) 0 N
   B) 25 N
   C) 50 N
   D) 100 N

3. A book rests on a table. Earth pulls down on the book, and the table pushes up on the book. Are these two forces an action-reaction pair? **[D]**

   A) Yes, because they are equal and opposite.
   B) Yes, because they both act on the book.
   C) No, because they are not equal in size.
   D) No, because they both act on the book, and an action-reaction pair acts on two different objects.

4. For the same book, what is the reaction force to "Earth pulls down on the book"? **[B]**

   A) The table pushes up on the book.
   B) The book pulls up on Earth.
   C) The book pushes down on the table.
   D) The air pushes up on the book.

5. A swimmer pushes the water backward with her arms. What force moves her forward? **[A]**

   A) The water pushes her forward.
   B) Her arms push her forward.
   C) Her inertia pushes her forward.
   D) Gravity pulls her forward.

6. A bug hits the windshield of a moving car. How do the forces compare? **[B]**

   A) The bug pushes on the car harder than the car pushes on the bug.
   B) The forces are equal in size.
   C) The car pushes on the bug harder than the bug pushes on the car.
   D) It depends on how fast the car is going.

7. In the same collision, the bug is splattered, but the car barely slows down. Why? **[A]**

   A) The bug's acceleration is much greater because it has much less mass.
   B) The car's acceleration is greater because it is moving faster.
   C) Their accelerations are equal because the forces are equal.
   D) The bug's acceleration is greater because the car exerts more force on it.

8. Two ice skaters, one 40 kg and one 80 kg, push off each other. Each skater feels a 120 N push. What is the acceleration of the 40 kg skater? **[D]**

   A) 0 m/s^2, because the forces cancel
   B) 1 m/s^2
   C) 1.5 m/s^2
   D) 3 m/s^2

9. A horse says, "The cart pulls back on me just as hard as I pull on it, so we can never move." What actually lets the horse and cart speed up? **[C]**

   A) The horse pulls on the cart harder than the cart pulls back.
   B) Nothing. The forces between the horse and the cart cancel, so they can't move.
   C) The ground pushes the horse forward harder than friction holds the cart back.
   D) The cart has no inertia once it is rolling.

10. An airplane heads north at 80 km/h while the wind blows it east at 60 km/h. What is the plane's speed relative to the ground? **[C]**

    A) 20 km/h
    B) 70 km/h
    C) 100 km/h
    D) 140 km/h

11. One rope pulls a box east with a force of 6 N, and another rope pulls it north with a force of 8 N. What is the net force on the box? **[B]**

    A) 2 N
    B) 10 N
    C) 10 m/s^2
    D) 14 N

12. Two forces, 6 N and 8 N, act on the same object. They can point in any directions. Which of these could NOT be the net force? [**D**]

    A) 2 N
    B) 10 N
    C) 14 N
    D) 16 N

13. A boat is aimed straight across a river. If the river's current becomes faster, how does that change the time it takes the boat to cross? [**A**]

    A) It takes the same time, but it lands farther downstream.
    B) It takes less time, because the current adds speed.
    C) It takes more time, because it travels farther.
    D) The boat cannot reach the other side.

Read this before answering 14–16. When a block sits on a ramp, its weight can be split into two components: one **perpendicular** to the ramp (pushing straight into the surface) and one **parallel** to the ramp (pointing down along the slope). The normal force from the ramp balances the perpendicular component. The parallel component is what makes the block want to slide.

14. A block rests on a tilted ramp. Compared to the block's weight, the normal force is: [**B**]

    A) zero
    B) less than its weight
    C) equal to its weight
    D) greater than its weight

15. The ramp is made steeper. What happens? [**D**]

    A) The normal force increases.
    B) Both components of the weight increase.
    C) Nothing changes, because the weight stays the same.
    D) The parallel component increases, and the normal force decreases.

16. A 20 N block is on a ramp. The ramp is tilted all the way until it is vertical, like a wall. What is the parallel component of the block's weight now? [**C**]

    A) 0 N
    B) 10 N
    C) 20 N
    D) 40 N

Read this before answering 17–19. When an object hangs from two ropes, the tension in each rope is a vector. The two tension vectors must add up to a resultant that points straight up and exactly balances the object's weight. Only the **vertical** components of the tensions help hold the object up.

17. A 100 N sign hangs from two vertical ropes that share the load equally. What is the tension in each rope? **[B]**

    A) 25 N
    B) 50 N
    C) 100 N
    D) 200 N

18. The same 100 N sign is rehung so the two ropes spread out at an angle, closer to horizontal. Now the tension in each rope is: **[D]**

    A) zero
    B) less than 50 N
    C) exactly 50 N
    D) more than 50 N

19. Why can a clothesline holding wet clothes never be pulled perfectly straight and horizontal? **[A]**

    A) A horizontal rope has no vertical component to hold the clothes up, so the tension would have to be impossibly large.
    B) The rope is too heavy to be straight.
    C) Gravity pulls the rope sideways.
    D) Tension only works in vertical ropes.

For 20–22, choose the law that best explains each situation.

20. Passengers on a bus lurch forward when the bus suddenly stops. **[A]**

    A) Newton's first law (inertia)
    B) Newton's second law (F = ma)
    C) Newton's third law (action-reaction)
    D) None of Newton's laws

21. The same push gives an empty shopping cart a greater acceleration than a full one. **[B]**

    A) Newton's first law (inertia)
    B) Newton's second law (F = ma)
    C) Newton's third law (action-reaction)
    D) None of Newton's laws

22. A rifle kicks backward into the shooter's shoulder when it is fired. **[C]**

    A) Newton's first law (inertia)
    B) Newton's second law (F = ma)
    C) Newton's third law (action-reaction)

D) None of Newton's laws

Part 2: Free Response

Show your work in the space given. Every number you write must have **units** (for example, 12 N or 3 m/s^2). When a blank asks for a **direction**, write one of the choices given in that question. Use g = 10 N/kg.

Reminder: when two vectors are at right angles, (resultant)^2 = (first)^2 + (second)^2.

Action-reaction and F = ma

23. A 60 kg student on roller skates faces a wall and pushes on it with a force of 120 N.
    a) How hard does the wall push on the student? **120 N**
    b) Direction of the wall's push (write "toward the wall" or "away from the wall"): **away from the wall**
    c) What is the student's acceleration? **2 m/s^2** (120 N / 60 kg)

24. Two skaters stand face to face and push off each other. Maya has a mass of 50 kg and Leo has a mass of 25 kg. Each pushes on the other with a force of 100 N. After the push, Maya moves east.
    a) What is the force on Leo? **100 N**
    b) Leo's direction (write "east" or "west"): **west**
    c) What is Maya's acceleration? **2 m/s^2** (100 N / 50 kg)
    d) What is Leo's acceleration? **4 m/s^2** (100 N / 25 kg)

25. A 1000 kg cannon fires a 10 kg cannonball. The cannon pushes the ball forward with a force of 5000 N.
    a) How hard does the ball push on the cannon? **5000 N**
    b) Direction of the ball's push on the cannon (write "forward" or "backward"): **backward**
    c) What is the cannonball's acceleration? **500 m/s^2** (5000 N / 10 kg)
    d) What is the cannon's acceleration? **5 m/s^2** (5000 N / 1000 kg)

Vectors

26. An airplane heads east at 120 km/h while the wind blows it north at 90 km/h.
    a) What is the plane's speed relative to the ground? **150 km/h** (120 squared + 90 squared = 22,500 = 150 squared)
    b) Is the plane's path closer to east or closer to north? (write "closer to east" or "closer to north"): **closer to east**

27. A boat is aimed straight across a river that is 40 m wide. The boat moves across at 4 m/s, and the current carries it downstream at 3 m/s.
    a) What is the boat's speed relative to the shore? **5 m/s** (3-4-5 triangle)

b) How long does it take the boat to cross? **10 s** (40 m / 4 m/s)
c) How far downstream does the boat land? **30 m** (3 m/s times 10 s)

28. A 3 kg box sits on frictionless ice. One rope pulls it east with a force of 9 N, and another pulls it north with a force of 12 N.

    a) What is the net force on the box? **15 N** (3-4-5 triangle times 3)
    b) What is the box's acceleration? **5 m/s^2** (15 N / 3 kg)

29. Two ropes pull on a box. One pulls east with a force of 6 N, and the other pulls west with a force of 8 N.

    a) What is the net force on the box? **2 N** (8 N - 6 N)
    b) Direction of the net force (write "east" or "west"): **west**
    c) If both ropes pulled east instead, what would the net force be? **14 N** (6 N + 8 N)

30. A ball is thrown with a speed of 10 m/s. The horizontal component of its velocity is 6 m/s. What is the vertical component of its velocity? **8 m/s** (6-8-10 triangle)

Forces on a ramp

31. A 50 N block rests on a level floor. What is the normal force on the block? **50 N**

32. The same 50 N block is placed on a frictionless ramp. Its weight splits into a perpendicular component of 40 N and a parallel component of 30 N.

    a) What is the normal force on the block? **40 N** (balances the perpendicular component)
    b) Direction of the normal force (write "into the ramp's surface" or "out of the ramp's surface"): **out of the ramp's surface**
    c) What is the net force on the block? **30 N** (the parallel component)
    d) Direction of the net force (write "up the ramp" or "down the ramp"): **down the ramp**
    e) What is the block's mass? **5 kg** (50 N / 10 N/kg)
    f) What is the block's acceleration? **6 m/s^2** (30 N / 5 kg)

Tension

33. An 80 N lamp hangs from two vertical ropes that share the load equally.

    a) What is the tension in each rope? **40 N** (80 N / 2)
    b) If one rope is cut and the lamp hangs from the other rope alone, what is the tension in that rope? **80 N**

34. A 120 N picture hangs from two ropes that make equal angles, like an upside-down V.

    a) Together, how much upward force must the two ropes provide? **120 N**
    b) What is the vertical component of the tension in each rope? **60 N** (120 N / 2)
    c) The horizontal component of the tension in each rope is 80 N. What is the tension in each rope? **100 N** (60 squared + 80 squared = 10,000 = 100 squared)